

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



231/1

Biology (Theory)

Paper 1

Time: 2 Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

1. Write your name, school and index number in the spaces provided above.
2. Write the date of the examination in the space provided above
3. Answer all questions in the spaces provided after each question only

FOR EXAMINER'S USE ONLY

QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORE
1 - 26	80	

Answer all questions in the spaces provided

1. Give the role of the following parts of microscope.

(3marks)

i) Mirror

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ii) Diaphragm

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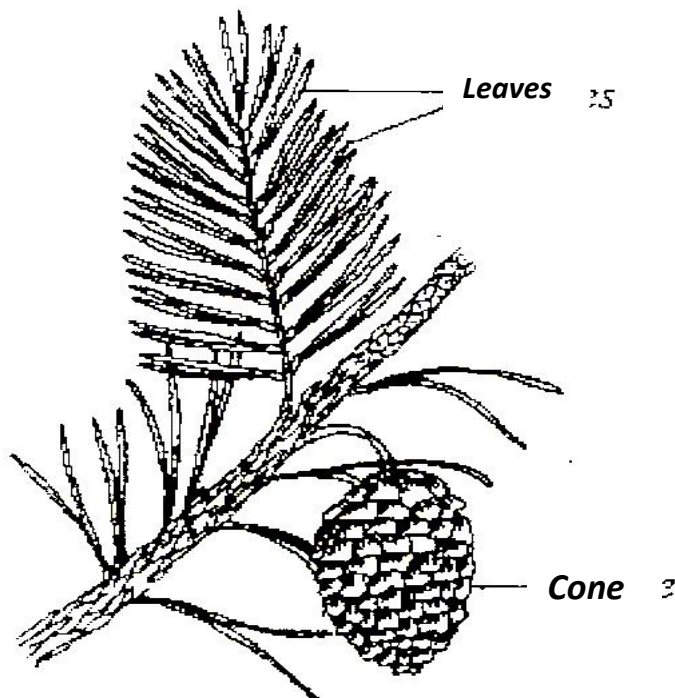
iii) Coarse adjustment knob

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2. (a) What **three** characteristics are used to divide the phylum arthropoda into classes? (3mks)

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(b) The diagram below represents a certain plant species.



i) State the class to which the plant belongs.

(1 mark)

.....

ii) State two differences between members of Gymnospermaphyta and Angiospermaphyta

(2 marks)

Gymnospermaphyta	Angiospermaphyta

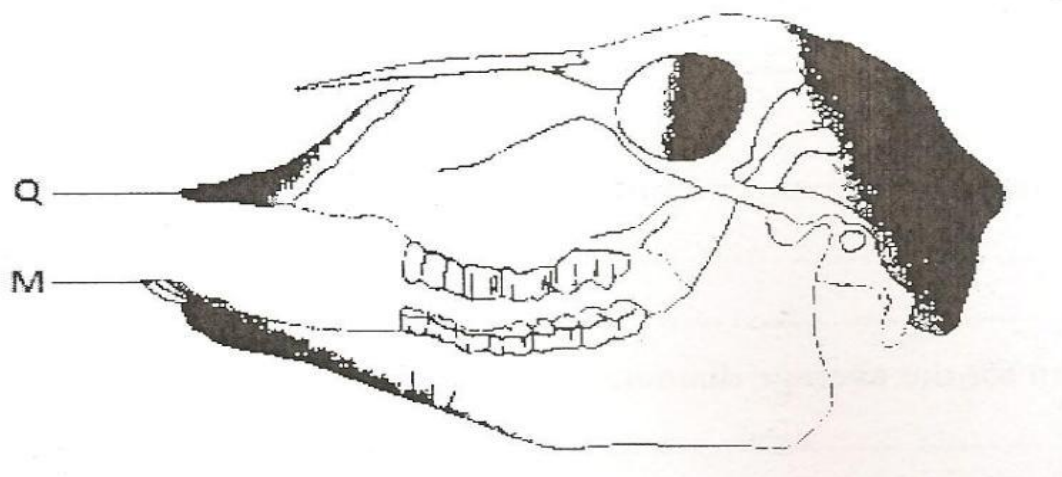
iii) State one observable xerophytic characteristic seen in the diagram above.

(1mark)

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.....

3. The following specimen was collected by form two students during a nature walk.



(a) Name the tooth labeled M.

(1 mark)

.....

(b) Name the part labeled Q and state its role.

(2 marks)

Name:

Role:

4. A solution of sugarcane was boiled with hydrochloric acid; sodium carbonate was added; cooled and benedict's solution was added then boiled. An orange precipitate was formed.

(a) Why was the solution boiled with hydrochloric acid? (1mark)

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(b) Why was sodium carbonate added? (1mark)

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(c) Name the type of reaction that takes place when simple sugars combine to form complex sugar.

(1mark)

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5. (a) Define the following terms as used in ecology. (2 marks)

i) Biosphere

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ii) Biomass

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(b) **Form** three students wanted to estimate the population of grasshoppers in 8km² grass field near a school compound. They captured 72 grasshoppers and marked them before returning them back to the field. After three days they made another catch of grasshoppers. They collected 147 grasshoppers of which only 29 had marks.

i) State why the second capture was done after three days. (1 mark)

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iii) From the data, calculate the population size of grasshoppers in the grass field. (2 marks)

6. a) State the most suitable biological tool for collecting the following organisms:-

i) A moth from a farm. (1 mark)

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ii) Ants from a tree trunk. (1 mark)

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7. The diagram below represents an organ from a finned bony fish. Study it and answer the question that follows



i. Identify the organ illustrated above. (1mark)

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.....

ii. State three adaptations of the part labeled S to its functions. (3 marks)

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8. a) State two characteristics that researchers select in breeding programs in plants(2marks)

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.....

b) In a maize the gene for purple colour is dominant to the gene for white colour. A pure breeding maize plant with purple grains was crossed with heterozygous plant. Using letter G to represent the gene for purple colour, work out the genotypic ratio of the offspring. (2 marks)

9. a) Name the organelle where Krebs's/ citric acid cycle take place in a cell. (1mark)

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b) In which organelle is each of the following found: (2mark)

i) Lytic enzymes

ii) Chromosomes

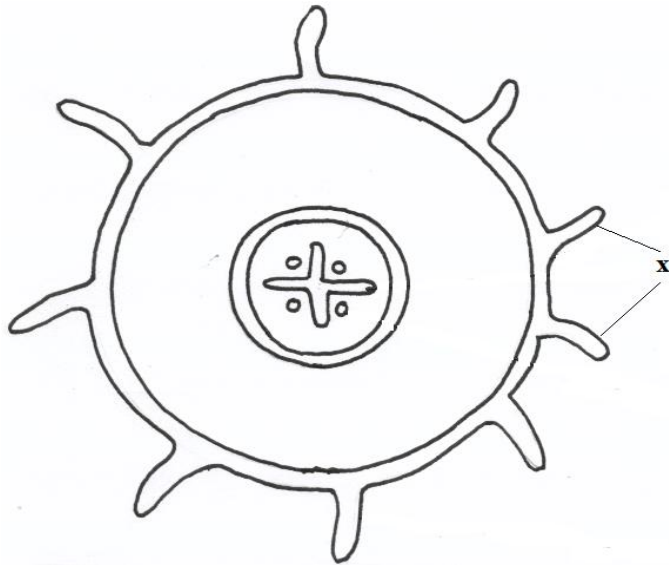
10. State the importance of osmo- regulation in organisms (2 marks)

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11. Explain why unicellular organisms such as paramecium lack complex organs for gaseous exchange. (2marks)

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12. The diagram below represents a transverse section of a plant part. Study it and answer the questions that follow.



a) Name the class in which the plant belongs. (1mark)

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b) Give a reason for answer (a) above. (1mark)

.....

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c) State **one** adaptation for the structures labeled X to their functions. (1mark)

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13. State the function of the following structures in the human ear.

(a) Semi – circular canals. (1mark)

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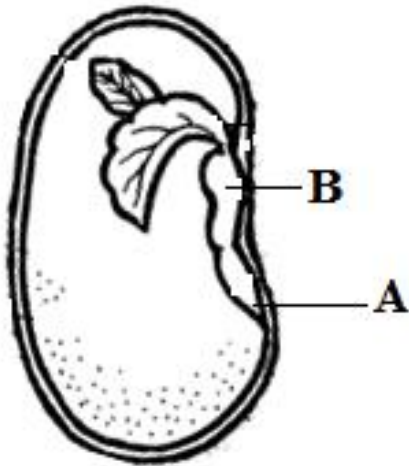
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(b) Eustachian tube. (1mark)

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14. The diagram below shows the internal structure of a broad bean seed. Study it and answer the questions that follow.



(a) Name the part labeled B. (1 mark)

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(b) Why is it important that the part labeled A develops first during germination? (2 mark)

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15. The diagram below represents a mammalian bone.



(a) Identify the bone (1mark)

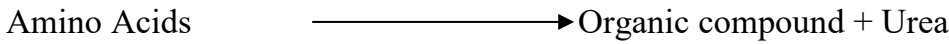
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(b) i) Name the bones that articulate with this bone at points K and L (2 marks)

K
.....

L
.....

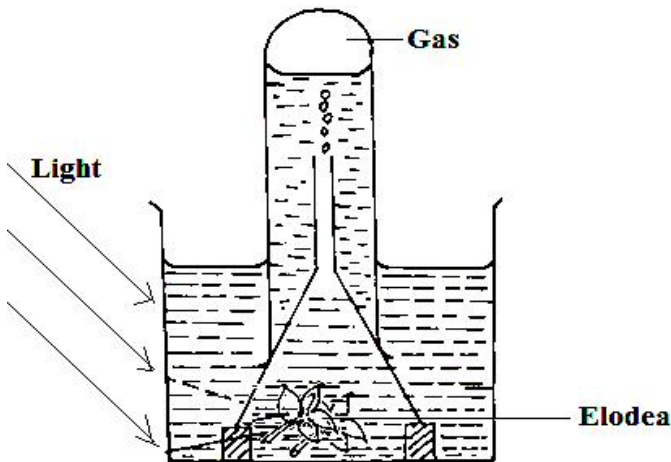
16. The equation below represents a metabolic process that occurs in the mammalian liver.



Name the process (1 mark)

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17. The diagram below represents a set up that was used to investigate a certain process in a plant.



(a) State the process that was being investigated. (1 mark)

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(b) Other than the factors shown, state two factors that would affect the process named in (a) above. (2 marks)

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18. Give an example of the sex linked trait in humans located on

Y – Chromosomes (1mark)

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X – Chromosomes (1 mark)

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19. Outline two roles of active transport in human beings. (2 marks)

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20. a). Name the causative organism for amoebic dysentery. (1 mark)

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b) State three preventive measures of schistosomiasis in human beings (3 marks)

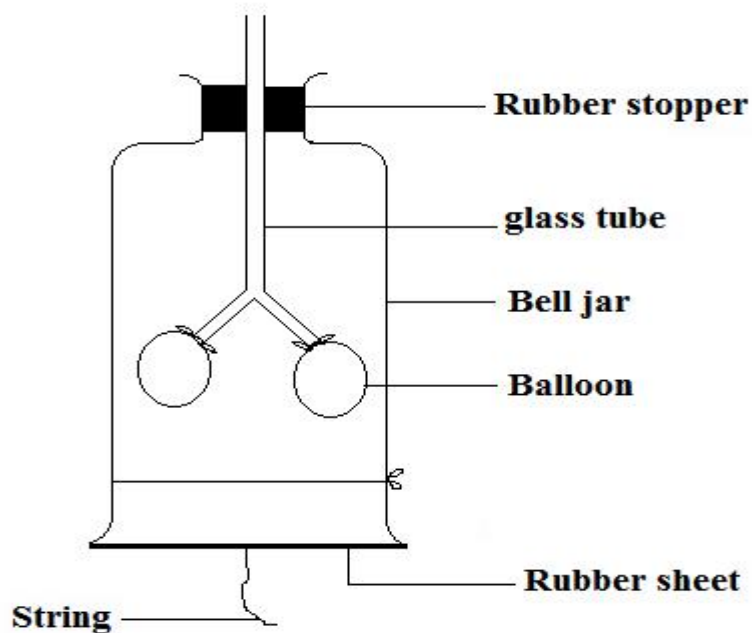
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21. Tom, a form two student set up the apparatus shown below to demonstrate the breathing mechanism in a mammal.



(a) What structure in a mammal is represented by each of the following? **(2marks)**

i) Glass tube.....

ii) Rubber sheet

(b) Explain what will happen to the balloons if the rubber is pulled down wards. **(2 marks)**

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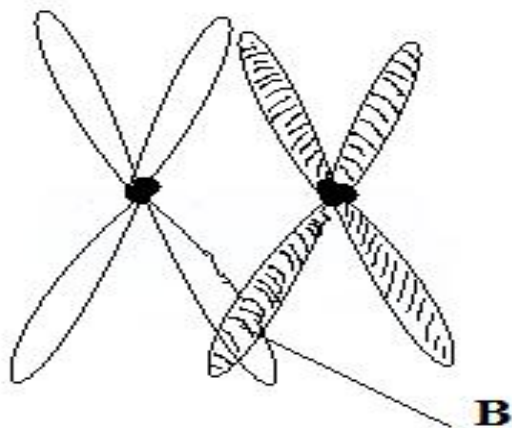
22. a) What is adaptive radiation? **(2marks)**

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b) Explain why crossbreeding is important in animal breeding. **(2 marks)**

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23. The diagram below shows a phenomenon which occurs during cell division.



a) Identify the stage of cell division in which this phenomenon occurs. (1 mark)

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b) State the importance of the phenomenon taking place in the part labeled B. (2 marks)

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24. a) State one advantage of double circulation over single circulation. (1mark)

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b) State two adaptations of blood capillaries to their functions. (2marks)

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25. a) Name a growth hormone that has inhibitory effects in plants growth. (1mark)

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b) State two characteristics of meristematic cells. (2marks)

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26. Write down two functions of exoskeleton. **(2 marks)**

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